Does Deterministic Generate–Validate–Repair Reduce Unsafe LLM-Generated Network Changes? A Confirmatory Paired Study

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***Abstract*—We preregistered and executed a sealed paired confirmatory study (n = 200 intent→configuration tasks) to test whether a deterministic generate–validate–repair (GVR) pipeline that consults a deterministic NetOps validator reduces validator-detected unsafe or semantically incorrect network changes relative to an LLM-only generation pipeline. The run used a single hosted model (gpt-5-mini) with adapter-enforced shared-initial generation semantics and a primary deterministic validator (deterministic_netops_validator_v5). Execution completed all planned pairs (completed_pairs = 200) consuming 200 model calls (one shared initial generation per task); no repair calls were issued because every initial candidate passed the validator. Paired contingency: n11 = 200, n10 = 0, n01 = 0, n00 = 0; estimate = 0.0; exact McNemar two-sided p = 1.0; 95% bootstrap percentile CI = [0.0, 0.0] (10,000 resamples, seed = 1729). Because the baseline validator-detected unsafe incidence was 0% on this task bank, the preregistered >=50% relative-reduction hypothesis could not be meaningfully evaluated. We report the sealed design, execution accounting (start: 2026-08-30T11:50:34.530059Z; end: 2026-08-30T12:12:07.541650Z), the Mininet 10% holdout (20 tasks) that produced zero validator–emulator disagreements, and an artifact-grounded discussion of operational implications and threats to validity (preregistration id: cnsms2026-gvr-effectiveness-02).**

I. Introduction

Automating intent→configuration translation with generative language models can increase NetOps productivity but risks producing incorrect or unsafe changes that cause outages or policy violations. Prior work therefore proposes grounding, deterministic validation, and repair loops (generate–validate–repair, GVR) to detect and correct unsafe candidates before application. The operational benefit of automated repair is conditional on three factors: (i) baseline prevalence of validator-detectable failures from the chosen generator/prompting policy; (ii) validator coverage and fidelity; and (iii) the available repair budget and repair-prompt effectiveness. We preregistered a narrowly scoped confirmatory question: Does an automated, deterministic GVR pipeline that consults a deterministic NetOps validator materially reduce validator-detected unsafe or semantically incorrect network changes relative to an LLM-only generation pipeline on a curated 200-task intent→configuration bank? The study was designed as a paired experiment that fixes the generator output per task and isolates the marginal effect of invoking a validator+repair on the same candidate.

II. Related Work

Recent literature documents three relevant threads for LLM-driven NetOps: (1) empirical characterizations of LLM factuality failures and mitigation patterns, (2) NetOps-specific prototypes that translate intent into configurations, and (3) system and threat analyses that focus on verification, guarded architectures, and attack surfaces in agentic tool use. First, cross-domain studies and surveys show that large language models produce factual errors, omissions, and hallucinations in operational tasks and that grounding, verifier-assisted repair, and uncertainty estimators can reduce but not eliminate such errors (e.g., Farquhar et al.; broad LLM surveys) [https://openalex.org/W4399803256, 10.1007/s11704-026-60308-3]. These works establish that mitigation effectiveness depends strongly on prompt design, sampling policy, retrieval/grounding fidelity, and task framing. Second, a growing set of NetOps prototypes demonstrates the feasibility of intent→configuration generation but also highlights evaluation gaps: NetConfEval and several intent-driven configuration works show that LLMs can synthesize network commands at prototype scale, yet reported evaluations often lack production-like instrumentation, end-to-end emulation, or preregistered inference procedures for failure-rate estimates [https://openalex.org/W4399629579, https://openalex.org/W4403635865, https://openalex.org/W4400072049]. Third, architectural proposals and empirical audits advocate combining generation with deterministic validators or simulators (generate–validate–repair, Batfish-style checks, and Mininet emulation) and propose bounded-autonomy guardrails to constrain unsafe agentic actions; at the same time, red-team and configuration-brittleness studies document that tool descriptions, chaining ambiguity, and persona/history can materially change generated actions and enable exploits if guardrails are incomplete [https://openalex.org/W4410125989, 10.36227/techrxiv.177004277.71795055/v1, 10.2139/ssrn.7203631, 10.21203/rs.3.rs-10646209/v1]. Complementary work on prompt sensitivity and LLM non-determinism shows that evaluation outcomes can shift with temperature, sampling, system prompts, or chain-of-thought strategies, which complicates comparisons across pipelines and motivates paired or shared-initial designs for controlled measurement

[https://openalex.org/W4402860127]. Finally, proposals for secure-by-default agent patterns and bounded-autonomy frameworks emphasize runtime veto layers, explicit tool registries, and staged validation to limit unsafe changes, but these remain largely proof-of-concepts whose real-world efficacy depends on validator coverage and emulation fidelity [10.36227/techrxiv.177006109.97957916/v1, 10.36227/techrxiv.177004277.71795055/v1]. Synthesizing these threads reveals a persistent evaluation gap: few studies provide preregistered, externally auditable measurements that isolate the marginal effect of deterministic repair on a fixed generator output under controlled sampling semantics. Our preregistered paired design addresses precisely this gap by enforcing shared-initial generation and deterministic validator checks, thereby separating generator correctness under a fixed prompt policy from any additional gains attributable to automated repair.

III. Methodology

Preregistration and inferential plan. We froze the experimental plan under preregistration id cnsms2026-gvr-effectiveness-02 before any model calls. The confirmatory design was paired and deterministic: n = 200 tasks (four topology clusters $\times$ 50 tasks), one shared deterministic initial LLM generation per task, and a repair budget of at most one automated repair call per task. The primary estimand was the paired_success_rate_difference_guarded_minus_baseline and the prespecified test was an exact two-sided McNemar test (α = 0.05). Confidence intervals for the paired difference were computed by the bootstrap percentile method with 10,000 resamples and bootstrap_seed = 1729. The analysis executor was paired_binary_analysis_v1 as recorded in the analysis artifacts.

Adapter, model, and sampling semantics. The run used a single hosted model declared as gpt-5-mini (provider label: openai_responses). Model configuration metadata (execution/model_configuration.json) records requested_model = "gpt-5-mini", max_output_tokens = 2000, maximum_attempts_per_call = 1, and temperature = null (deterministic sampling enforced by the adapter). Crucially, the hosted_netops_gvr_v1 adapter enforced shared_initial generation semantics: for each task the system generated exactly one initial candidate which was deterministically reused for both the baseline (LLM-only) and guarded (GVR) conditions. This shared-initial design eliminates per-condition sampling variance and isolates the incremental effect of invoking validation and an allowed repair step on the identical generator output. Adapter accounting in deterministic_reconciliation.json documents hosted_model_call_event_count = 200, stage_counts = {shared_initial_generation: 200}, and model_calls_used = 200, confirming the enforced single shared invocation per task and that no separate condition-specific sampling occurred.

Task bank, validators, and holdout emulator. The task bank is a curated, machine-checkable intent→configuration set (200 tasks) in which each task encodes initial and expected final states plus workflow policies (e.g., make_before_break_uplink_switchover, must_be_down_for_fields). The primary deterministic checker (deterministic_netops_validator_v5, validator_version = "5.0") runs syntactic parsing, intent/constraint enforcement, and structured diagnostics (fields available in validator traces include valid flag, violation_count, violation_codes, parsed_command_count, required_command_count, and difficulty annotations). A Mininet-based data-plane emulator was preregistered as a secondary disagreement detector applied to a randomized 10% holdout (20 tasks) to detect validator–emulator mismatches; the executed contamination_summary artifact reports zero disagreement flags for that holdout. The preregistration specified deterministic treatment rules: pairs flagged for validator–emulator disagreement would be excluded from the primary analysis per the contamination_plan; sensitivity analyses for alternative treatments were preregistered but not invoked because no flags occurred.

Repair policy, budgeting, and missingness handling. Per the preregistration, when the primary validator reported violations the pipeline was permitted to issue up to one automated repair prompt to the same LLM for that task (maximum repair calls across the sample = 200). Repair prompts, if used, would be captured in repair_provider_trace fields in the episode artifacts; none appear in the archived episodes because no repairs were applied. Missingness and exclusion rules were deterministic: pairs where neither condition completed because of infrastructure failures would be excluded and logged; pairs with contamination flags (validator–emulator disagreement) would be excluded from the primary confirmatory test. The execution manifest and missingness_summary show complete_pair_count = 200 and zero failed episodes, so the planned missingness handling was not invoked in practice.

Execution accounting and artifact capture. The autonomous run executed between 2026-08-30T11:50:34.530059Z and 2026-08-30T12:12:07.541650Z. The adapter enforced the preregistered maximum model calls (maximum_model_calls = 400) and per-task caps (initial generation = 1 call; repair calls <= 1). The run consumed 200 model calls (one shared initial generation per task), with cache_miss_count = 200 and cache_hit_count = 0 as recorded in the deterministic reconciliation artifacts. All model calls, prompts, raw responses, validator traces, provider traces, and analysis artifacts (execution/raw_results.jsonl, analysis/paired_contingency_table.csv, analysis/condition_summary.csv, analysis/contamination_summary.csv, analysis/analysis_log.jsonl, execution/model_configuration.json) were archived; representative per-task artifacts (e.g., execution/scoring/task-000001-baseline.json) show validator_trace fields and scoring outcomes. The analysis pipeline executed the exact McNemar test and bootstrap CI generation exactly as preregistered; analysis/analysis_log.jsonl records bootstrap_resamples = 10000 and bootstrap_seed = 1729.

Design rationale and implications for internal validity. The

shared_initial paired design was chosen to isolate the pure marginal effect of validation+repair on a fixed generator output; this reduces variance attributable to stochastic sampling and makes the confirmatory test focused on whether repair can convert a validator-failing candidate into a validator-passing one for the same initial output. The trade-off is explicit: the design sacrifices generality across multiple independent generator draws for stronger internal control. Preregistration, deterministic adapter semantics, and frozen stopping rules were used to avoid post-hoc analytic choices. Deterministic reconciliation artifacts confirm pair_accounting_consistent = true and all_deterministic_consistency_checks_passed = true.

Limitations baked into the methodology. The methodology intentionally restricts scope: a single model, a deterministic prompt/sampling policy, a one-repair budget, and a curated task bank. These constraints reflect an explicit confirmatory question about marginal GVR effect under a tightly controlled generator policy rather than an evaluation of alternative sampling, multi-step repair strategies, or cross-model generality. When interpreting outcomes, readers should account for these preregistered methodological choices documented in the archived plan.

IV. Results

Execution outcomes. The run completed all planned episodes: completed_episode_count = 400 (shared initial generation reused across conditions) and complete_pair_count = 200. Adapter accounting shows model_calls_used = 200 (one shared initial generation per task) and cache_miss_count = 200. No repair calls were issued because every initial candidate passed deterministic_netops_validator_v5. Condition summaries (analysis/condition_summary.csv) show baseline successes = 200/200 and guarded successes = 200/200 (success_rate = 1.0 for both). The paired contingency (analysis/paired_contingency_table.csv) is n11 = 200, n10 = 0, n01 = 0, n00 = 0.

Primary inferential analysis. With zero discordant pairs the observed paired difference estimate = 0.0. The exact McNemar two-sided p = 1.0. The bootstrap percentile 95% CI (10,000 resamples; seed = 1729) is [0.0, 0.0]. Under the preregistered estimand and test the prespecified >=50% relative-reduction hypothesis is logically untestable on this sample because there were no baseline validator failures to reduce.

Representative diagnostics. Representative per-task artifacts (for example, pair-000001) confirm that the shared initial generator outputs matched the task required_sequence and that validator_trace.validation_after.valid = true with violation_count = 0. Difficulty annotations in validator traces (examples: make_before_break_uplink_switchover, paired_link_atomic_mtu_change) and parsed_command_count/required_command_count show that tasks exercised medium-to-very_hard workflow constraints despite generating validator-satisfying candidates. The Mininet 10% holdout (20 tasks) executed as preregistered; analysis/contamination_summary.csv reports zero validator–emulator disagreements, and deterministic reconciliation artifacts report all consistency checks passed (pair_accounting_consistent = true).

Unexecuted branches. Planned sensitivity branches for flagged contamination treatments and alternative missingness handling were not invoked because no tasks were flagged and no episodes failed. Secondary exploratory estimands (average repair iterations, GVR-introduced failure rate, model-call cost per successful repair) are empty in the analysis bundle because no repair attempts occurred; the analysis artifacts record these secondary_results as empty.

V. Discussion

Conditional interpretation. The degenerate null outcome demonstrates a logical constraint: when the chosen generator, prompt template, and sampling policy produce validator-passing candidates for every item in a workload, an added validator-coupled repair step cannot reduce validator-detected failures. This observation is artifact-grounded and operationally important: the marginal value of automated repair depends on baseline validator failure prevalence and on validator coverage.

Why baseline failures were zero (artifact-based explanations). First, the prompt template and enforced shared_initial semantics produced outputs that conformed to explicit task-bank constraints. Second, the task bank is curated with clear required_sequences and machine-checkable constraints that map directly to validator checks—this reduces semantic ambiguity and makes correct outputs easier to generate with a deterministic prompt policy. Third, the primary validator enforces the task-encoded constraints but may omit device-level idiosyncrasies detectable only by an end-to-end emulator or real device; the recorded Mininet holdout produced zero disagreements but the run also discloses a missing labeled selection seed for that holdout (see reproducibility note). Fourth, the single repair budget and shared_initial design remove per-condition re-sampling as a pathway to obtain alternative repairable candidates.

Operational implications. Before deploying automated repair, practitioners should measure baseline validator failure prevalence on representative operational workloads and examine validator coverage with end-to-end emulation. When baseline validator failures are rare under the chosen generator/prompt policy, investing in richer validator fidelity (vendor semantics, device models), multi-sample generation, multi-model ensembles, or targeted adversarial testing may yield higher safety returns than automated repair. Conversely, when baseline failures are non-negligible, a well-engineered GVR pipeline with sufficient repair budget can reduce validator-detected issues; realized gains will depend on repair-prompt quality, budget, and validator completeness.

Threats to validity. Internal validity: preregistration, deterministic adapter enforcement, and frozen stopping rules minimize post-hoc choices, but the shared_initial design intentionally trades sampling generality for a controlled paired comparison. Construct validity: the primary validator enforces

explicit task constraints but may not model all real device behaviors; emulator disagreements would reveal such blind spots but none were observed on the 20-task Mininet holdout. External validity: a single hosted model (gpt-5-mini), single prompt/sampling policy (shared initial deterministic sampling), a curated synthetic/public task bank, and a single repair budget were used; results may differ with other models, nonzero temperature sampling, multiple samples per condition, adversarial workloads, or production device heterogeneity. Conclusion validity: with zero discordant pairs the McNemar test is uninformative for the preregistered effect size; nonetheless the observed zero-discordance is a reproducible operational datum documented in the archived artifacts.

Reproducibility note and documented gap. All execution artifacts (model-call logs, validator traces, paired contingency tables, bootstrap parameters) are archived in the execution bundle. The Mininet holdout evidence is captured in analysis/contamination_summary.csv which reports zero disagreements for the 20-task holdout; this artifact demonstrates that the secondary emulator check executed and found no validator–emulator mismatches. A remaining reproducibility gap is that the explicit randomized selection seed for the Mininet holdout is not labeled in the archived artifacts, preventing exact deterministic reconstruction of which tasks were drawn into the holdout despite the contamination summary reporting zero disagreements. We disclose this gap transparently in the Disclosure Statement.

VI. Limitations

- Task-bank scope: the confirmatory sample used a curated synthetic/public intent→configuration bank (n = 200); results therefore reflect this bank and may not generalize to broader production workloads or adversarial distributions.
- Single-model and shared-initial setting: only gpt-5-mini (hosted) with adapter-enforced shared-initial generation was tested (execution artifacts record hosted_model_call_event_count = 200 and stage_counts shared_initial_generation = 200). Outcomes may differ across model families, independent per-condition sampling, nonzero temperature sampling, or larger repair budgets.
- Validator coverage and oracle limitations: the primary deterministic validator enforces a defined set of syntax/intent constraints; validators can fail to capture real device idiosyncrasies, vendor-specific behaviors, or emergent failure modes detectable only by end-to-end deployment.
- Adversarial robustness untested: this confirmatory run did not include active adversarial injection or red-team tool-description attacks; prior audits indicate such vectors can bypass naive defenses and remain a separate concern.
- Reproducibility gap for contamination check: the randomized holdout selection seed for the Mininet contamination check is absent from the labeled artifacts, preventing exact reconstruction of the holdout selection.
- Single repair budget: the GVR pipeline allowed at most one automated repair prompt per task; multi-step repair strategies, different repair budgets, or human-in-the-loop approvals were not exercised here.

VII. Conclusion

We preregistered and executed a sealed paired study comparing LLM-only generation to a deterministic GVR pipeline on 200 NetOps intent→configuration tasks using gpt-5-mini and deterministic_netops_validator_v5. Both pipelines produced validator-passing configurations for every task (n11 = 200; no discordant pairs), resulting in an observed paired difference of 0.0, exact McNemar p = 1.0, and bootstrap 95% CI = [0.0, 0.0]. The preregistered >=50% relative-reduction hypothesis could not be evaluated because baseline validator failures were absent. The artifact-grounded outcome clarifies that GVR's marginal value is workload dependent: when baseline validator-detectable failures are rare, automated repair yields no measured benefit for those validator-detected errors. Future studies should target workloads with nontrivial baseline failure rates, expand validator fidelity with richer emulation or real-device checks, evaluate multi-sample and multi-model policies, and explore multi-step or human-in-the-loop repair budgets to characterize practical operational gains. All execution artifacts and analysis outputs are archived for independent inspection under the preregistration id cited above.

Disclosure Statement

Disclosure (mandatory). This study executed under the autonomous post-lock preregistration cnsms2026-gvr-effectiveness-02. Declared model: gpt-5-mini (provider label: openai_responses) invoked via the hosted_netops_gvr_v1 adapter. The exact initial master prompt is archived at provenance/master_prompt.txt (SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196 add77f5acdf1582bb39b15ba). Topic constraints: the human Research Director limited the study to Generative AI and LLMs for NetOps focusing on safety/reliability of generated network changes. Frameworks and tools: orchestration used hosted_netops_gvr_v1 and the deterministic task generator deterministic_netops_task_generator_v5. Primary deterministic validator: deterministic_netops_validator_v5 (intent/constraint checks). A Mininet-based emulator was preregistered and executed as a secondary disagreement detector on a randomized 10% holdout (20 tasks); the artifact analysis/contamination_summary.csv shows zero disagreements for that holdout. Preregistration and freeze: the experimental plan (estimand, sample size n = 200, paired design, stopping rules, max model calls = 400, repair budget = 1 per task, shared-initial generation semantics) was frozen before any model calls. Autonomous execution and artifacts: the run executed autonomously under the frozen plan (start: 2026-08-30T11:50:34.530059Z; end: 2026-08-30T12:12:07.541650Z); model calls, prompts, seeds, validator traces, emulator traces, logs, and the artifact manifest are archived in the execution bundle. Model-call

budget and usage: 200 model calls were used (one shared initial generation per task); up to 200 repair calls were allowed but none were applied because initial candidates passed the validator. Human involvement: the human Research Director selected the topic family and pre-lock constraints; no human manual editing or selective rewriting of technical manuscript content occurred after the autonomy lock. Deviations and restarts: no post-preregistration design changes or technical restarts occurred. Known reproducibility gap: the explicit randomized selection seed used to draw the Mininet 10% holdout is not present as a labeled artifact in the archive; this absence prevents exact deterministic reconstruction of the drawn holdout despite the contamination summary reporting no disagreements. The master prompt archive reference above is the immutable prompt required by the track.

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